

Cover letter — Journal of Cheminformatics

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6 October 2026

To the Editors, *Journal of Cheminformatics*

Dear Editors,

I am pleased to submit “**Ensembles Across Model Families Outperform Any Single Architecture: A Single-Configuration, Leakage-Controlled Benchmark of 31 Molecular Property Models Across 44 Datasets**” for consideration as a **Research article**.

QSARena is an open-source, MIT-licensed, code-free and command-line benchmarking pipeline that evaluates 31 models — conventional machine learning, gradient boosting, deep tabular and graph neural networks, 3D pretrained models and ensembles — across 44 molecular property-prediction datasets from five public suites (TDC, MoleculeNet, Polaris, PODUAM and ChemML) under a single fixed, leakage-controlled configuration.

The contribution is methodological rather than a new state-of-the-art model, and I have framed it that way. That trees and descriptor-based models remain competitive with deep and pretrained architectures is by now a well-replicated result, and the manuscript cites the recent cross-suite literature to that effect. What this paper adds is a uniform, single-configuration benchmark applied across five suites with train-only feature selection, and — its central result — a decomposition of the gap between test-selected and cross-validation-selected leaderboard standing into two separable causes: the breadth of the candidate model library and the cost of honest, held-out selection. Because both protocols derive from one run and one reference set, the decomposition is less sensitive to the known quality problems of published leaderboards than any absolute rank. The manuscript also quantifies how much cross-validation optimism nested, train-only feature selection removes.

The work fits the journal’s stated commitment to publishing benchmarking studies and to full reproducibility, and it is in the scope of the journal’s recent collection on evaluating AI and machine learning models in cheminformatics. All code is openly available under an OSI-approved license at github.com/ScottCoffin/QSARena, a versioned archive including the per-molecule prediction files will be deposited at Zenodo before publication with its DOI cited in the manuscript, and the manuscript regenerates every reported number from the deposited artifacts via an automated check.

I have stated the principal limitation prominently: all results derive from a single split and seed per dataset, so conclusions that rest on close margins are presented as provisional. The conclusions I emphasize — the breadth of the winner distribution, the one-sided selection effect and the order-of-magnitude cost differences — do not depend on those margins.

This manuscript is original, has not been published previously, and is not under consideration elsewhere. The author declares no competing interests. The work was supported by the California Office of Environmental Health Hazard Assessment, and computation used NSF ACCESS Jetstream2 under allocation CIS261142. Use of large language model assistance is documented in the Methods (Section 2.13), as the journal requires.

Thank you for considering this submission.

Sincerely,

Scott Coffin

California Office of Environmental Health Hazard Assessment, Sacramento, CA, USA
scott.l.coffin@gmail.com